Analysis of NBS: Co-benefits

CUNY EES 79903

Seminar on Nature-Based Solutions

Brett Branco and Kyle McKay

October 6, 2026

Lecture Overview:

- 415-420 – Housekeeping: field trip, projects
- 420-445 – Lecture: assessing outcomes
- 445-500 – Case study: Bronx River restoration planning
- 500-615 – Discussion: González-García et al. (2025), Sadaf and Bruck (2026), Chrysoulakis et al. (2021)

	Leader #1	Leader #2
~~Sep 8: NBS fundamentals~~	~~Grace~~	~~Jesse~~
~~Sep 15: NBS fundamentals~~	~~Michelle~~	~~Shahela~~
~~Sep 22: Structured decision making and community engagement~~	~~Jesse~~	~~Aurelia~~
~~Sep 29: Analysis – ecological~~	~~Carol~~	~~Julia~~
~~Oct 6: Analysis – co-benefits~~	~~Shahela~~	~~Aurelia~~
Oct 20: Analysis – engr performance	Grace	Michelle
Oct 27: Analysis – engr performance	Abdellah	Julia
Nov 3: Alternatives analysis	Aurelia	Abdellah
Nov 10: Performance-based design	Aaryan	Michelle
Nov 17: Constructability	Grace	Aaryan
Nov 24: Performance monitoring	Shahela	Jesse
Dec 1: Performance monitoring	Julia	Carol
Dec 8: Life cycle analysis	Aaryan	Abdellah
Dec 15: Course synthesis	Kyle	Brett

Field trip planning

October 11th – 900am-300pm

- 845 – Meet at Brooklyn College
- 900 – VAN LEAVES
- 945-1045 – Rockaways – Beach 43rd St. / USACE project
- 1130-100 – Broad Channel – Lunch, West Pond, marsh islands, Sunset Cove marsh, thin layer placement sites
- 100-230 – Shirley Chisolm State Park
- 300 – Adjourn from Brooklyn College

Project Reminder

- Oct 20 – Recommended timeframe to compile initial references
- Nov 3 – Recommended timeframe for outlining ideas
- Nov 17 – Initial submission of traceable account
- Nov 24 – Review of traceable account by classmates
- Dec 8 – Final submission of traceable account
- Dec 15 – Course synthesis (McKay)

Topic	First option
Qualitative foundations of project planning	Jesse
Facilitation, engagement, and teaming	Aurelia
Ecological analysis of NBS	Julia
Economic analysis of NBS	
Co-benefits analysis of NBS	
Engineering analysis of NBS	Abdellah
Alternatives analysis and structured decision making	
Performance-based design	Aaryan
Constructability	Michelle
Performance monitoring	
Project establishment	
Asset management	Grace

Thinking about benefits and costs

Why do we need to assess benefits and costs?

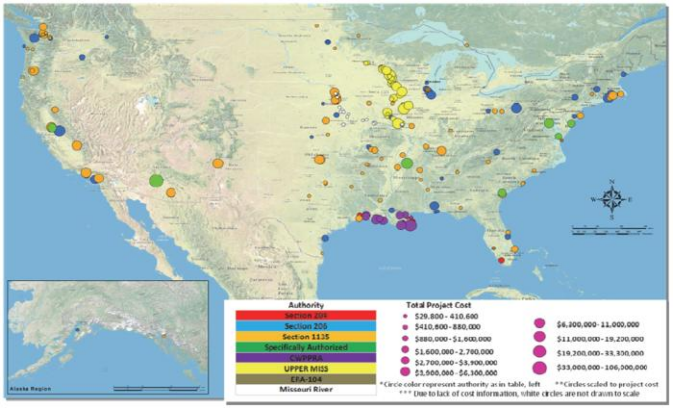
Programmatic Budgeting and Reporting **Project Decision-Making** ⟷ **Project Narrative Building**

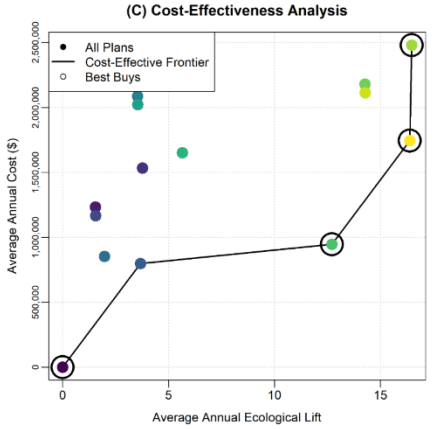
Performance Criteria	Maximum Score	Description
Habitat Scarcity	25	Regional or national scarcity of the habitat type
Connectivity	25	Extent to which organismal movement is improved or a critical component is added to increase biodiversity
Special Species Status	10	Provides significant contribution to a key life requisite in the potential range of a listed species
Hydrologic Character	20	Refers to the timing, magnitude, duration, frequency, and rates of change of aquatic systems
Geomorphic Condition	20	Establishment of suitable structure and set of physical processes for successful restoration
Self-sustaining	20	Applies to advanced phases, goal of project to be self-sustaining
Plan Recognition	10	Recognizes Corps ecosystem restoration projects that contribute to watershed or basin plans as emphasized in the "Civil Works Strategic Plan"
TOTAL	130	Maximum potential score

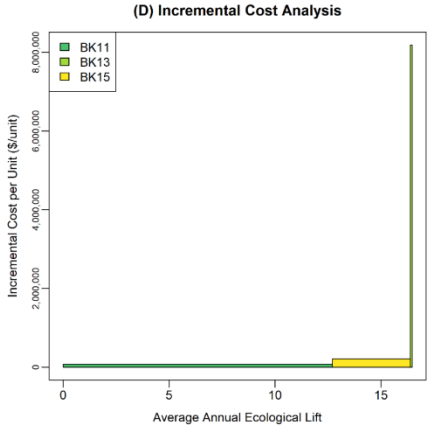


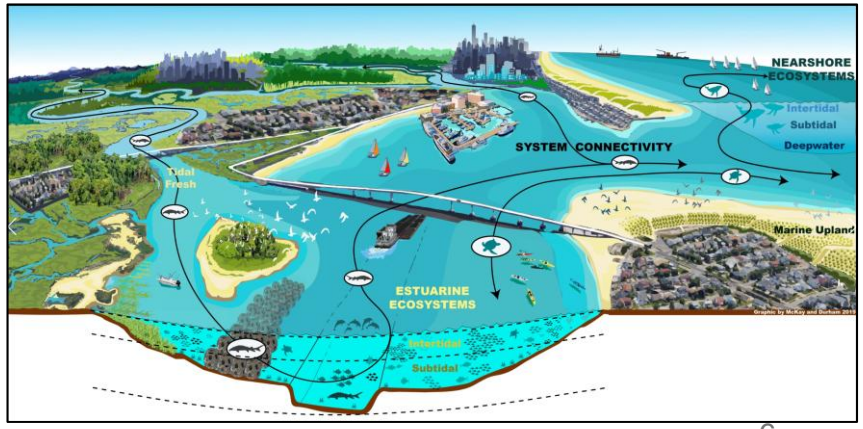




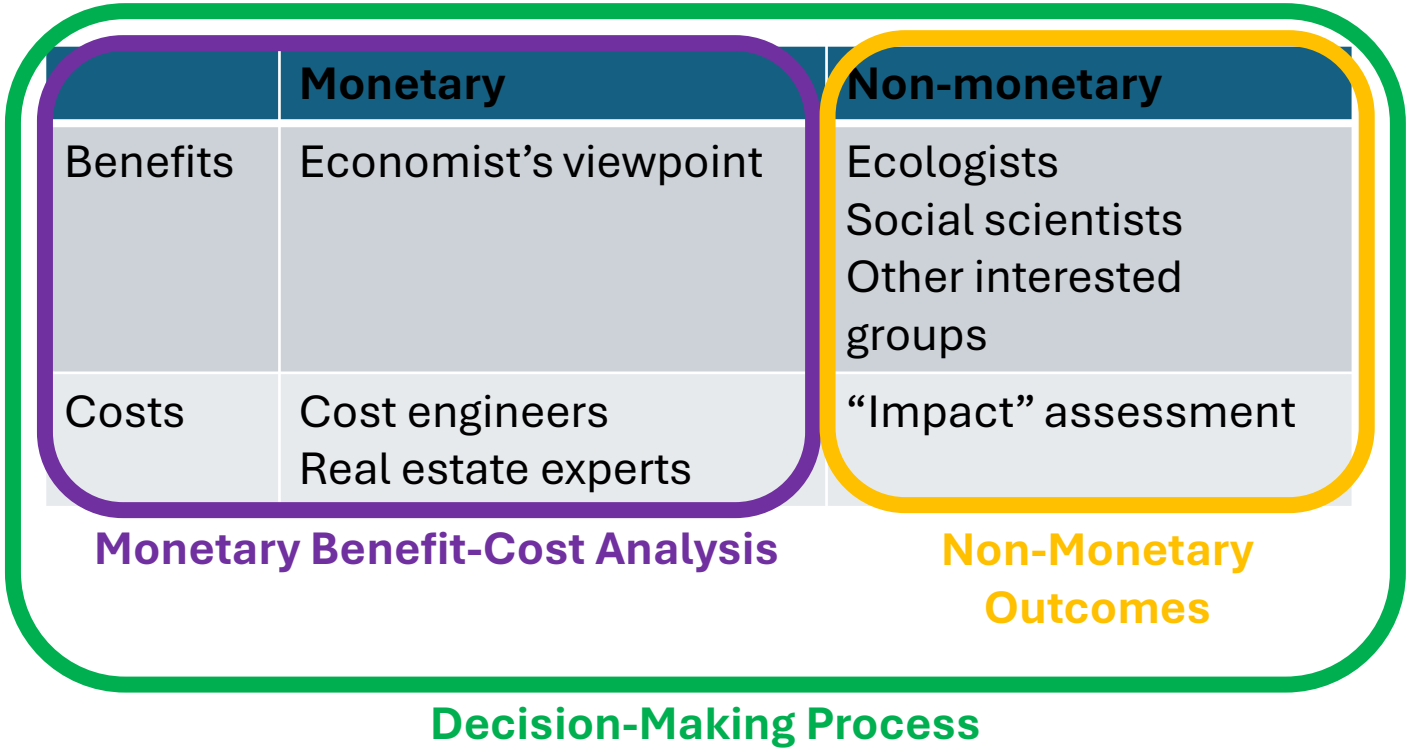








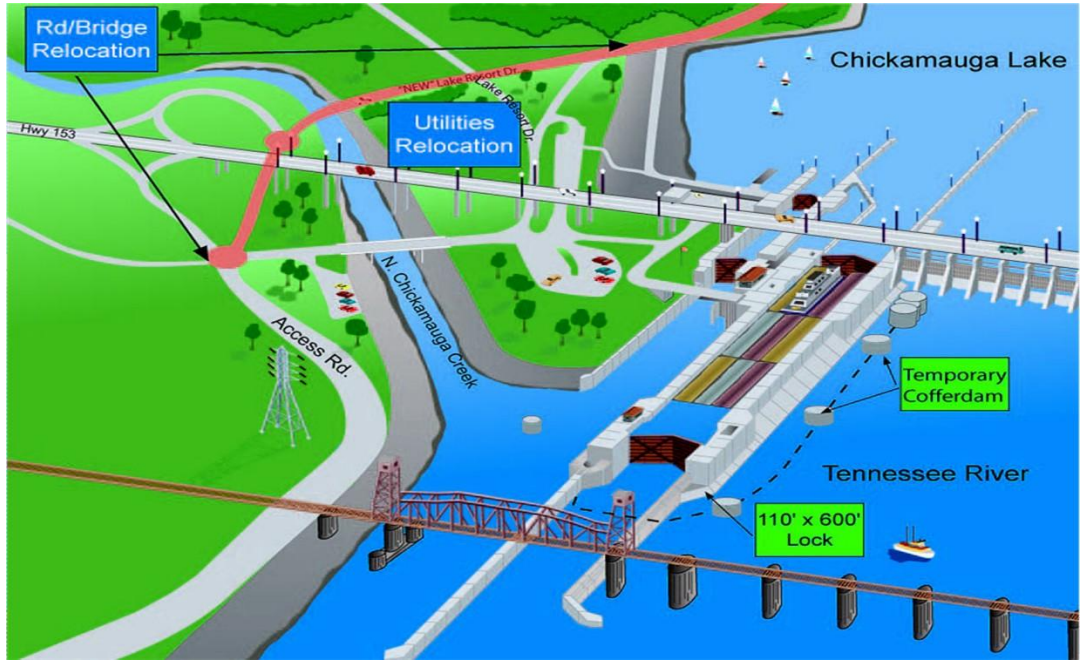
A basic structure to talk about benefits and costs



Monetary Benefit - Cost Analysis

Costs

- Land Purchase
- Dam Construction
- Dam Operation
- Power lines
- Irrigation systems
- Navigation aids



Benefits

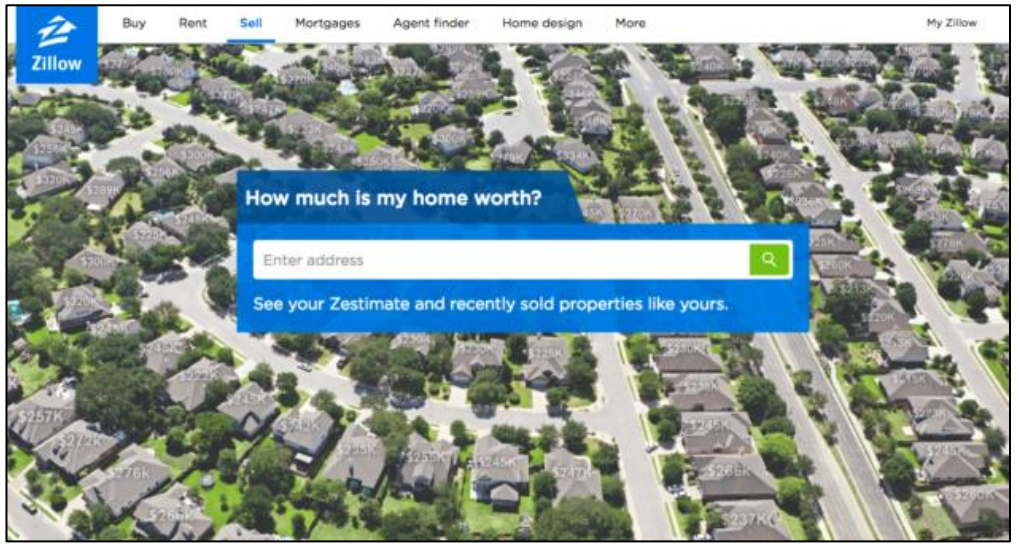
- Cheaper electricity
- Fewer floods
- More irrigation water
- More recreation
- Easier navigation
- Increased property values

Monetary Benefits

- Two types of economic benefits:
  - Increasing the monetary benefit of good things (e.g., hydropower)
  - Decreasing the monetary cost of bad things (e.g., avoided flood damage)
- Are all benefits the same?
  - Is the benefit adding to the overall size of the economy (e.g., a new source of recreation)?
  - Is the benefit trading-off one part of the economy for another (e.g., reservoir vs. river recreation associated with dam removal)?
- How are benefits distributed through time?

Monetary Costs

- Capital costs incurred: real estate, construction
- Indirect costs incurred: infrastructure relocation, water rights
- Uncertainty in costs: “contingency”
- Operations and maintenance
- Monitoring and adaptive management





Economic benefit-cost analysis

- Benefit - Cost Ratio (BCR):
  - Ratio of Benefits to Costs: r = B / C
  - BCR > 1 means more benefits than costs
- Net Value:
  - Benefits minus costs: NV = B - C
  - NV > 0 means more benefits than costs
- Internal Rate of Return (irr):
  - Discount rate that makes B(irr) = C
  - irr > market rate is a good investment
  - “Time value of money”

Costs ($)	Benefits ($)
Land Purchase	Recreational kayaking
Project Planning	Reduced water treatment
Construction	Increased property value
Monitoring	Recreational fishing

Challenges to overcome:

- Can we monetize all of the outcomes?
- What’s included? Capital, operations, long-term sunsetting?
- What are we leaving out?
- Are values and money the same thing?

Examples of common benefit-cost metrics

- Costs = $1, Benefits = $2
  - Great ratio (B/C=2), but not much profit (B-C=$1)
  - Good for small, private investor

- Costs = $1,000, Benefits = $1,100
  - Poorer ratio (1.1), but more profit ($100)
  - Good for larger investor, like the government

- Costs = $1,000, Benefits = $1,500 in 20 yrs
  - Internal rate of return (irr) = 2 %, which is not a great investment
  - Alternative investments would yield better returns

Economic benefit-cost analysis

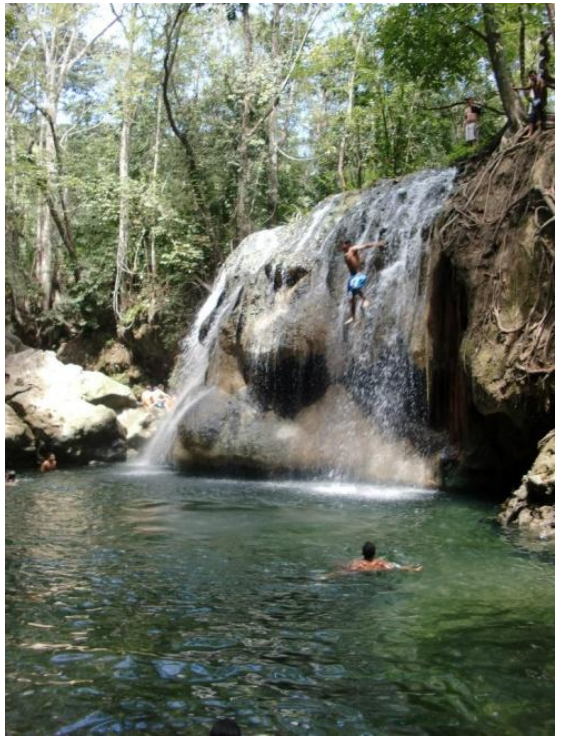
Alternative	Capital Cost ($)	Operations Cost ($/yr)	Economic Benefit ($)	Benefit-Cost Ratio	Net Benefits ($)
Do nothing	0	0	0	n/a	0
A	1,000,000	**3,000**	9,000,000	9	8,000,000
B	2,000,000	10,000	20,000,000	**10**	18,000,000
C	10,000,000	100,000	50,000,000	5	**40,000,000**

Some examples of non-monetary benefits

- Aesthetics and preferences
- Recreation – boating, fishing, swimming
- Passive – wildlife watching, peace and quiet
- Spiritual values – baptism, seclusion
- Commercial values
- Ecological outcomes









Some examples of non-monetary costs

- Non-monetary costs are largely the domain of regulatory policies (i.e., they are often “externalities”)

- Environmental damage
- Lost access to cultural or historical resources
- Occupational and safety hazards associated with building a project
- Environmental justice
- Equity of benefits and costs
- …

Other ways to think about benefits and costs: Structured decision making

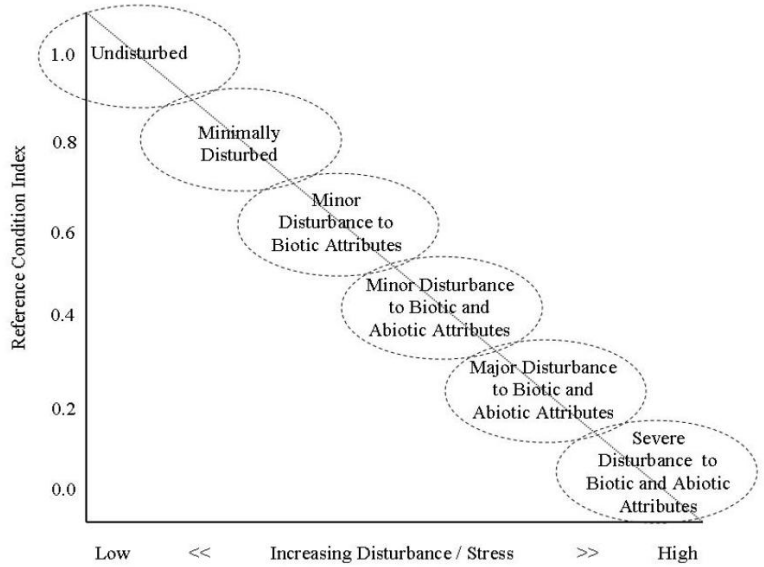
Define the **PR**oblem
Set **O**bjectives
Develop **A**lternatives
Assess **C**onsequences
Make **T**rade-offs



Figure: National Conservation Training Center (NCTC)

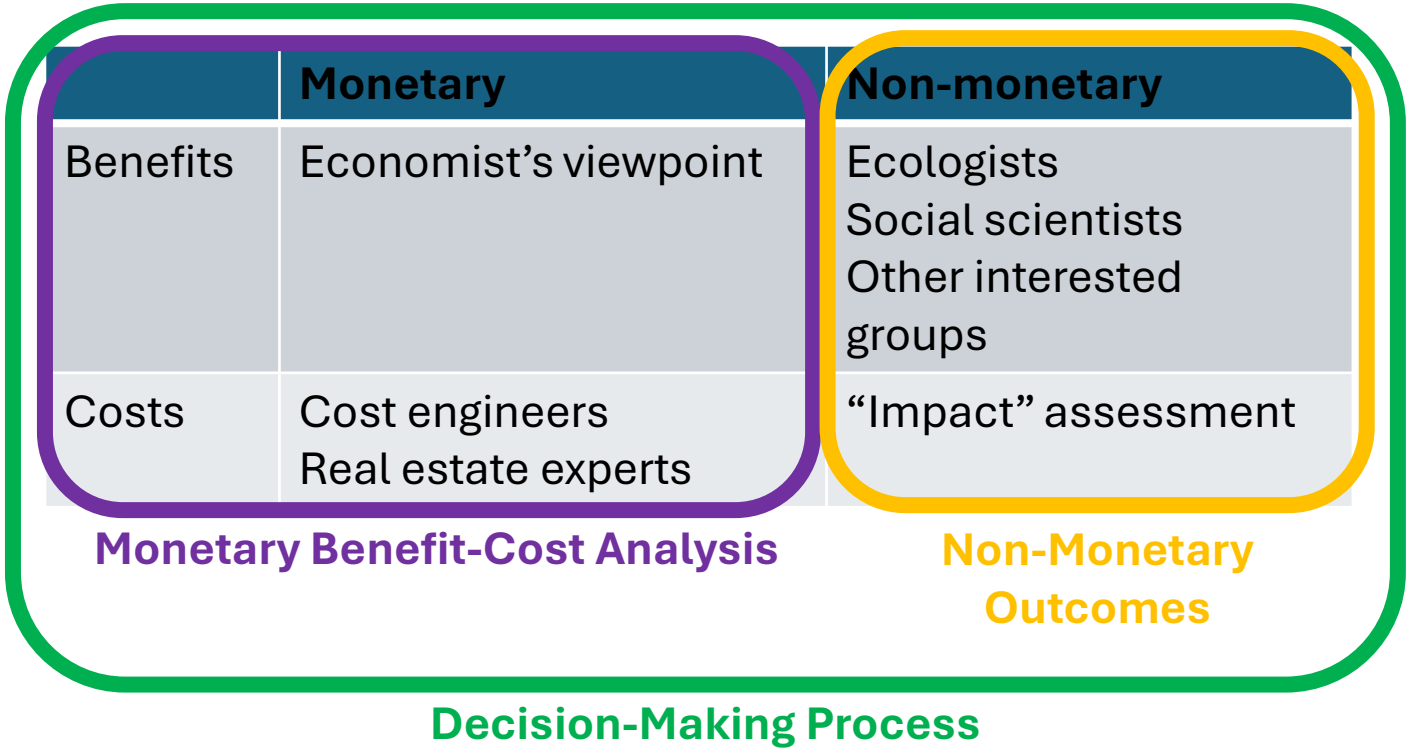
Other ways to think about benefits and costs: Reference conditions

- Assesses directionality and relative benefit
  - 10% improvement, 60% of reference
- Great for comparing multiple actions
  - Option-A is definitely better than Option-B
- Not so great for quantitative reporting
  - Does a 10% improvement “fix” the problem?
  - What is the net economic benefit of this improvement?



- How do we create a benchmark (i.e., a “reference”) to measure success against?
  - Historical conditions (pre-Columbian, 1940 when the USGS opened, last year), minimally disturbed, least disturbed, best attainable, regional example
  - Do we set the bar high or low?

A basic structure to talk about benefits and costs

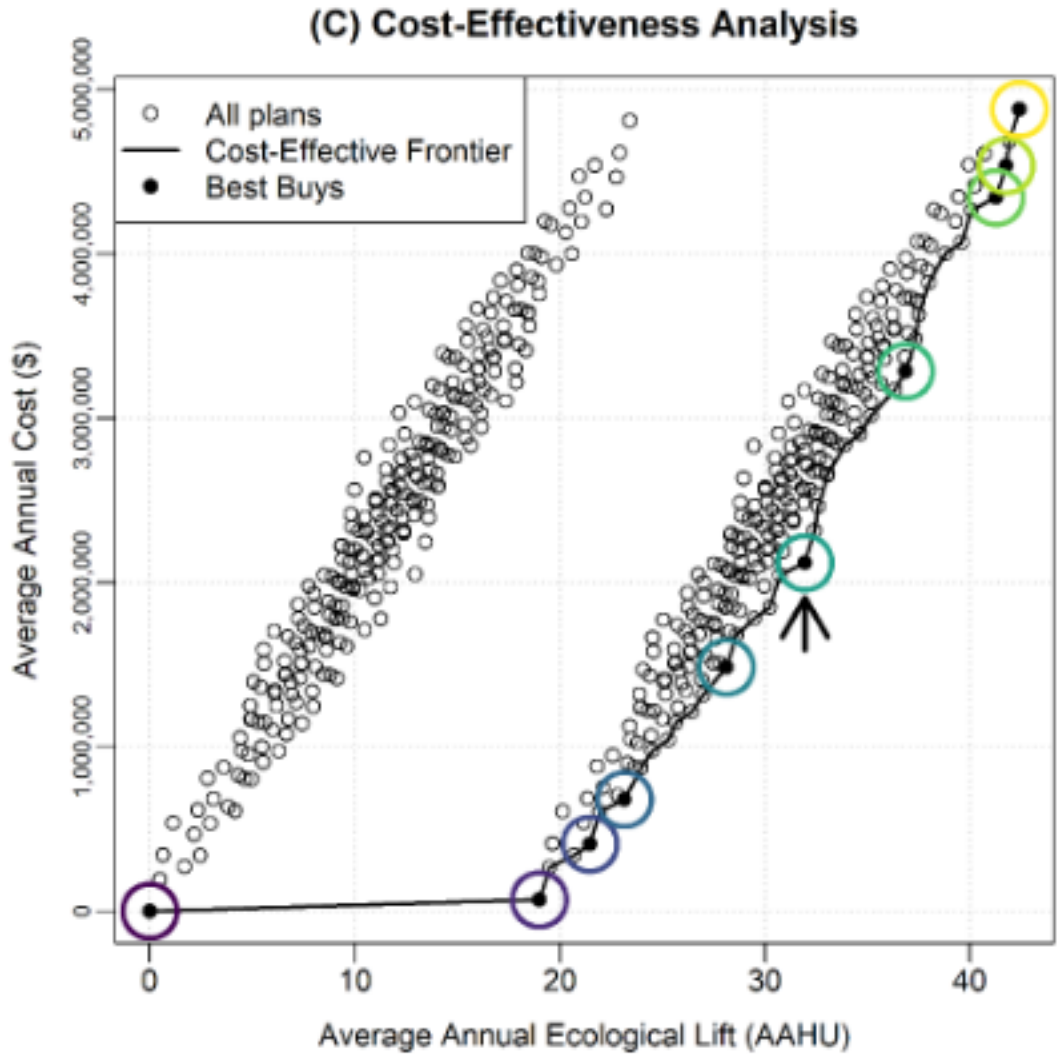


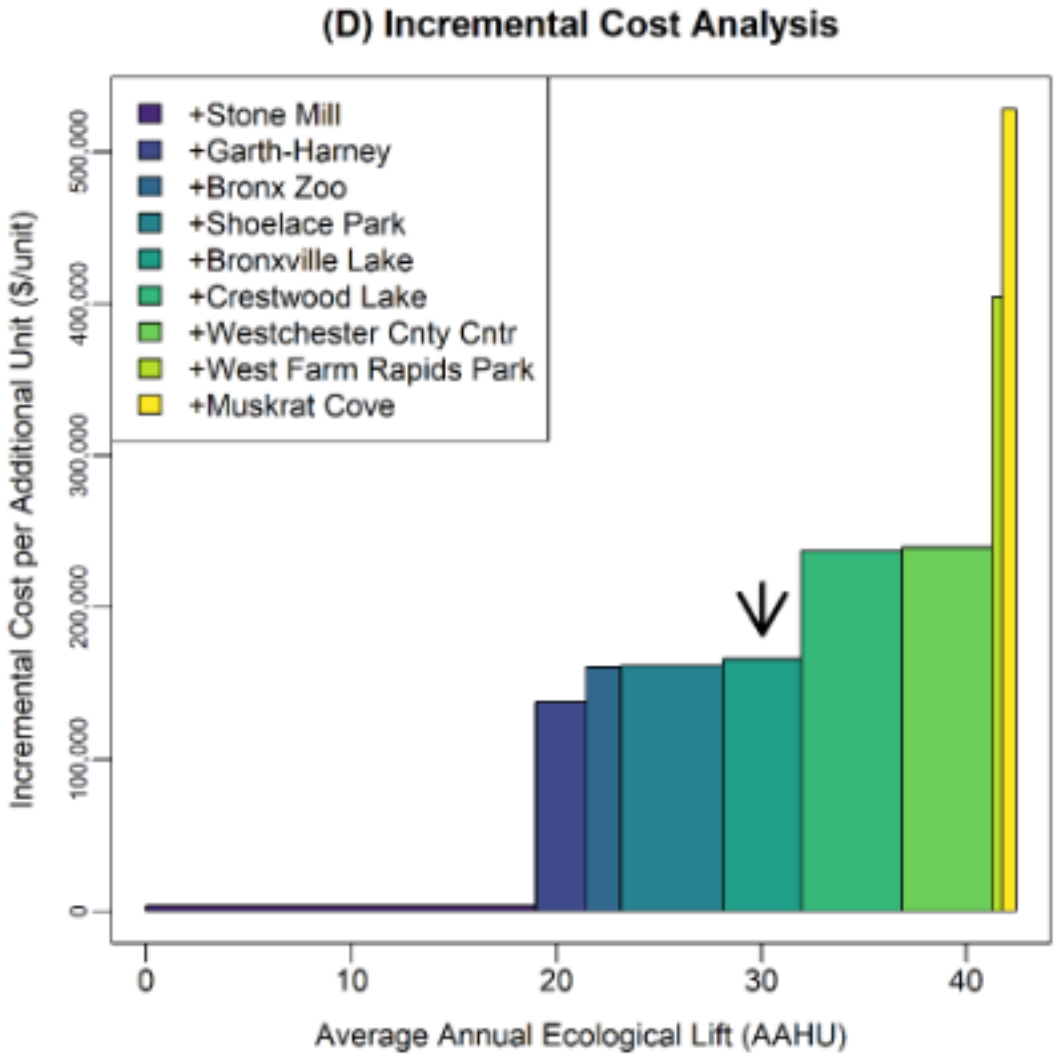
Bronx River Restoration, Hudson-Raritan Ecosystem, New York



Hudson-Raritan Estuary Restoration Team: Lisa Baron, Peter Weppler, Carissa Scarpa, Diana Kohtio, Danielle Tomasso, Mike Morgan, Gail Wooley, Maya Dehner

Conventional view of restoration outcomes





Bronx River Restoration Prioritization

- Which sites are the most “important”?
- Are factors beyond ecological benefits and cost at play?
- How can we communicate important outcomes to decision-makers and the public?



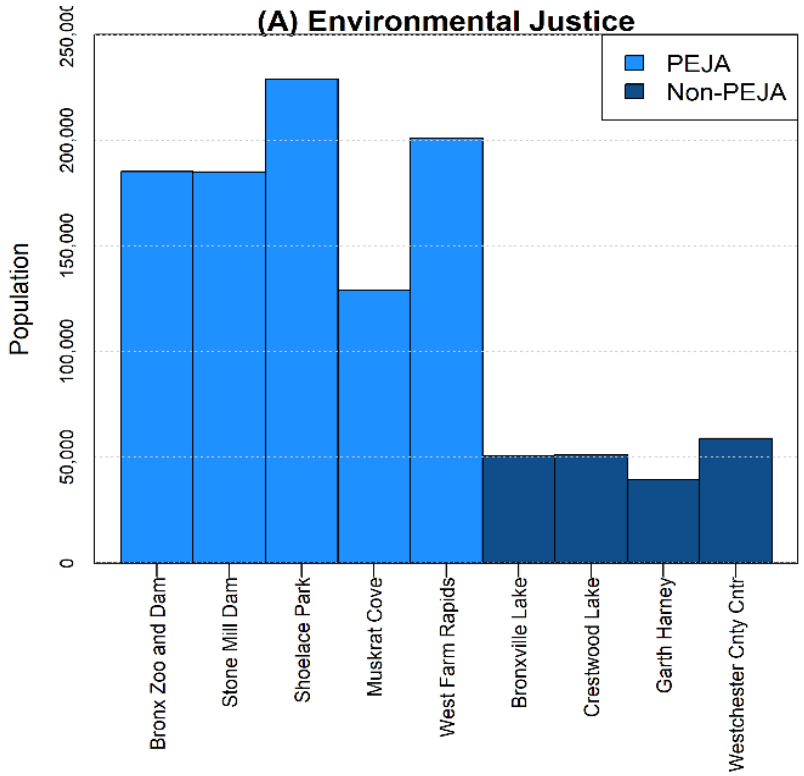
Collecting “secondary” criteria to contextualize decision-making

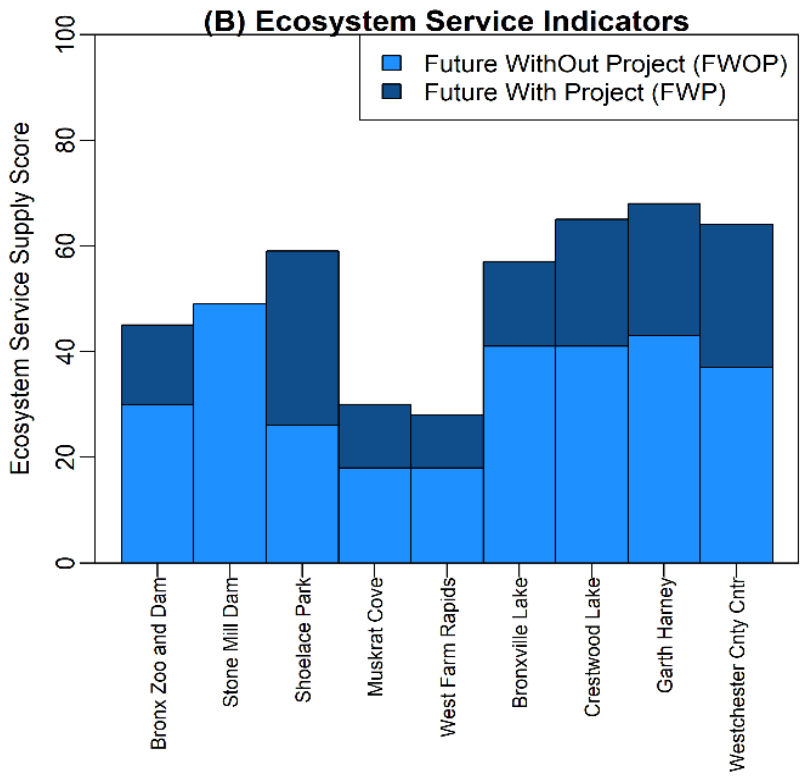
- *Social equity*: Potential Environmental Justice Areas
- *Ecosystem services*: Qualitative scoring system
- *Stakeholder support*: Perceived support and number of cost-share partners

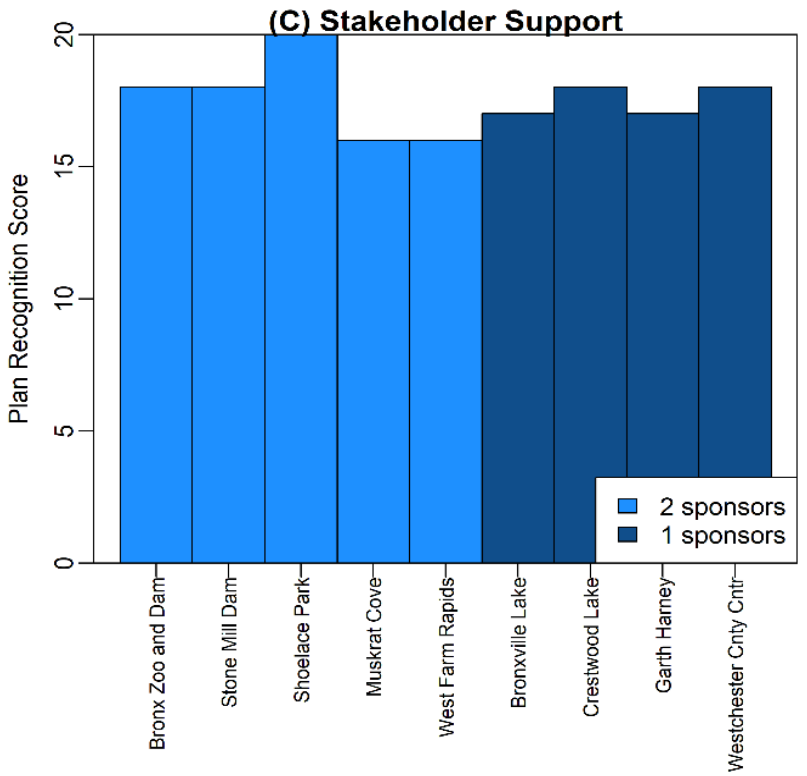
Category	Optimal 20 19 18 17 16	Suboptimal 15 14 13 12 11	Marginal 10 9 8 7 6	Poor 5 4 3 2 1 0	No Action	Future w/Project
Ecosystem Services – Benefit Relevant Indicators						
Flood risk management	Measurable reduction in flood levels significantly reducing risk for residents and damages	Clear qualitative reduction in the flood risk	Minor, site-specific reductions in flood risk outcomes	Negligible or zero effect on flood risk outcomes	2	10
Intersection with inland navigation	High proximity to navigation projects. Site actively performs a functional role for the navigation mission (e.g., sediment disposal).	High proximity to navigation projects. Site indirectly improves the navigation mission (e.g., sediment retention).	Low proximity to navigation projects. No obvious current connection to navigation activities.	Active interference with navigation projects or mission.	6	6
Recreational opportunities	Meets all four objectives from the CRP (2009): direct access, indirect access, vistas, upland access routes	Meets 3 of 4 categorical objectives from the CRP	Meets 2 of 4 categorical objectives from the CRP	Meets 1 of 4 categorical objectives from the CRP (or zero objectives)	9	16
Air & thermal regulation	Project area is 75-100% vegetated	Project area is 50-75% vegetated	Project area is 25-50% vegetated	Project area is 0-25% vegetated	7	14
Improve water quality	Significant, quantitative contribution to sediment settling, nutrient uptake, and carbon retention	Major, qualitative contribution to sediment settling, nutrient uptake, and carbon retention	Minor, indirect benefits to sediment settling, nutrient uptake, and carbon retention	Little (if any) effect on sediment settling, nutrient uptake, and carbon retention outcomes	2	13



Summarizing data for decision-makers







Summarizing data for decision-makers

Watershed-Scale Plans	Lift (AAHU)	Cost ($)	Unit Cost ($/AAHU)	Number of PEJAs	Total Ecosystem Services Score	Total Population	Stakeholder Priority Sites
No action	0.0	0	0	0	0	0	0
Stone Mill Dam only	19.0	70,508	3,711	1	0	185,029	1
All prior actions + Garth-Harney	21.5	409,640	19,089	1	25	224,348	1
All prior actions + Bronx Zoo and Dam	23.2	681,874	29,452	2	40	409,410	2
All prior actions + Shoelace Park	28.1	1,486,280	52,847	3	73	638,280	3
All prior actions + Bronxville Lake*	31.9	2,121,092	66,390	3	89	689,018	3
All prior actions + Crestwood Lake	36.9	3,287,524	89,164	3	113	740,328	4
All prior actions + Westchester County Center	41.3	4,342,402	105,203	3	140	799,159	5
All prior actions + West Farm Rapids Park	41.8	4,537,760	108,663	4	150	999,991	5
All prior actions + Muskrat Cove	42.4	4,880,878	115,089	5	162	1,129,102	5

Paper Discussion

González-García, A., Palomo, I., Codemo, A., Rodeghiero, M., Dubo, T., Vallet, A. and Lavorel, S., 2025. Co-benefits of nature-based solutions exceed the costs of implementation. Cell Reports Sustainability, 2(3).

Sadaf, A. and Bruck, J., 2026. Optimizing shoreline suitability models framework: integrating advanced GIS methodologies at Aberdeen Proving Ground, Maryland. Frontiers in Environmental Science, 14, p.1812979.

Chrysoulakis, N., Somarakis, G., Stagakis, S., Mitraka, Z., Wong, M.S. and Ho, H.C., 2021. Monitoring and evaluating nature-based solutions implementation in urban areas by means of earth observation. Remote Sensing, 13(8), p.1503.

González-García et al. (2025)

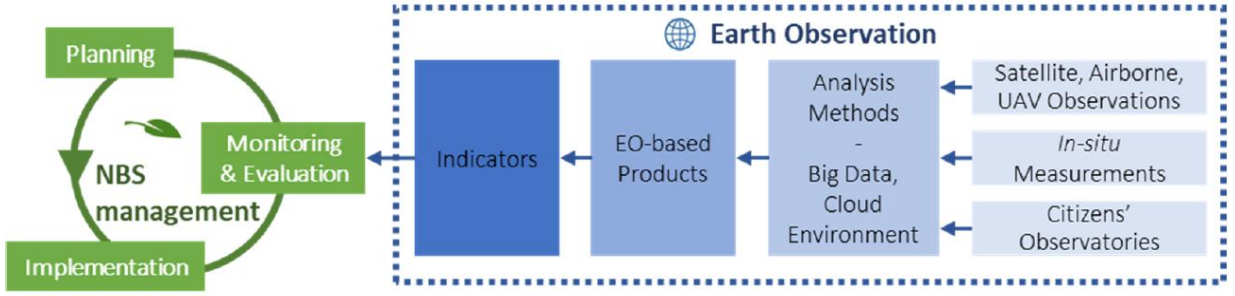
- The paper addresses biodiversity outcomes, but leaves them out of the summary (Table 1). How is this information used to inform decision making?
- What ecosystem services were not included? Why?
- Is there a spectrum of ecosystem service metrics ranging from proxies to monetization?
- Do forests provide more ecosystem services?
- Are benefit-cost ratios for different NBS types helpful or harmful?

Sadaf and Bruck (2026)

- What about the military installation changes the planning context?
- Should models be “standardized” or project specific?
- Is there circular logic in using models to calibrate models?
- What is the difference in categorical and continuous variables?

Chrysoulakis et al. (2021)

- What is earth observation?
- Tasked vs. untasked satellites
- What are the strengths and weakness of remote sensing?
- How do these methods align with NBS monitoring goals?
- How do these methods align with urban monitoring goals?



Satellite / System	Spatial Resolution	Temporal Resolution (Revisit)	Primary Applications	Cloud Sensitivity
Landsat	30m	16 days	Land cover, historical trends	High
Sentinel-2	10m	5 days	Agriculture, urban mapping	High
MODIS	250m–1km	Daily	Climate, large ecosystems	Moderate/High
Sentinel-1	Varies (~10–20m)	6–12 days	Floods, soil moisture, radar	None (Radar)
Commercial (e.g., Planet)	Sub-meter–3m	Daily / Near real-time	Asset tracking, tactical details	High (Optical)

General Discussion

- How do these modeling approaches differ, contrast, or complement each other?
  - Transparency and replicability
  - Directness in alignment
  - Depth of on-the-ground information
  - Philosophical approach to modeling
- Making trade-offs in assessment methods
  - What are your constraints? Resources like time, funding, expertise
  - What are the objectives? Alignment with goals and directness
  - How do I “tell the story”? Analysis paralysis vs. under-quantification

Next Week = NO CLASS

Oct 20th papers (McKay is out):

Bredes, A., Tso, G., Gittman, R.K., Narayan, S., Tomiczek, T., Miller, J.K. and Morris, R.L., 2024. A 20-year systematic review of wave dissipation by soft and hybrid nature-based solutions (NbS). Ecological Engineering, 209, p.107418.

Corwin, E., Reguero, B.G., McKay, S.K., Piercy, C.D., Pontee, N.I., Tomiczek, T., Jameson, T., Narayan, S., Sridharan, V.K. and Bledsoe, B.P., 2026. Nature-Based Design Standards: Past, Present, and Future. ASCE OPEN: Multidisciplinary Journal of Civil Engineering, 4(1), p.04026002.

Esquivel-Muelbert, J.R., Fontoura, L., Zawada, K., Erickson, K., Figueira, W., Madin, J.S. and Bishop, M.J., 2026. The natural architecture of oyster reefs maximizes recruit survival. Nature, pp.1-5.